

# HOMEWORK SOLUTIONS

## PROBLEM 1

$$(i) \quad f(x) = \begin{cases} x + \pi & -\pi < x < 0 \\ -x + \pi & 0 < x < \pi \end{cases}$$

Area of a triangle

$$d_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \left( \frac{1}{2} (2\pi)(\pi) \right) = \frac{\pi}{2}$$

$$d_n = \frac{1}{2\pi} \left[ \underbrace{\int_{-\pi}^0 (x+\pi) e^{-inx} dx}_{(1)} + \underbrace{\int_0^{\pi} (-x+\pi) e^{-inx} dx}_{(2)} \right]$$

$$(1) \quad \int_{-\pi}^0 (x+\pi) e^{-inx} dx$$

$$u = x + \pi \\ du = dx$$

$$v = -\frac{1}{in} e^{-inx} \\ dv = e^{-inx} dx$$

$$= -\frac{1}{in} (x+\pi) e^{-inx} + \frac{1}{in} \int e^{-inx} dx$$

$$= -\frac{1}{in} (x+\pi) e^{-inx} + \frac{1}{n^2} e^{-inx} \Big|_{-\pi}^0$$

$$= \left( -\frac{1}{in} \pi + \frac{1}{n^2} \right) - \left( 0 + \frac{1}{n^2} e^{i\pi n} \right) = -\frac{\pi}{in} + \frac{1}{n^2} - \frac{(-1)^n}{n^2}$$

$$(2) \quad \int_0^{\pi} (-x+\pi) e^{-inx} dx$$

$$u = -x + \pi \\ du = -dx$$

$$v = -\frac{1}{in} e^{-inx} \\ dv = e^{-inx} dx$$

$$= -\frac{1}{in} (-x+\pi) e^{-inx} - \frac{1}{in} \int e^{-inx} dx$$

$$= -\frac{1}{in} (-x+\pi) e^{-inx} - \frac{1}{n^2} e^{-inx} \Big|_0^{\pi} =$$

$$= \left( 0 - \frac{1}{n^2} e^{-i\pi n} \right) - \left( -\frac{\pi}{in} - \frac{1}{n^2} \right) = \frac{\pi}{in} + \frac{1}{n^2} - \frac{(-1)^n}{n^2}$$

$$d_n = \frac{1}{2\pi} \left[ -\frac{\pi}{in} + \frac{1}{n^2} - \frac{(-1)^n}{n^2} + \frac{\pi}{in} + \frac{1}{n^2} - \frac{(-1)^n}{n^2} \right] = \frac{1}{2\pi} \left[ \frac{2}{n^2} - \frac{2(-1)^n}{n^2} \right] \\ = \frac{1}{\pi n^2} (1 - (-1)^n)$$

$$f(x) = \frac{\pi}{2} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1}{\pi n^2} (1 - (-1)^n) e^{inx} \quad \text{or} \quad f(x) = \frac{\pi}{2} + \sum_{n \text{ odd}} \frac{2}{\pi n^2} e^{inx}$$

# PROBLEM 1

$$(ii) \quad f(x) = \begin{cases} 0 & -\pi < x < 0 \\ 1 & 0 < x < \pi \end{cases}$$

$$d_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} (\pi) = \frac{1}{2}$$

$$d_n = \frac{1}{2\pi} \left[ \int_0^{\pi} e^{-inx} dx \right] = \frac{1}{2\pi} \left[ -\frac{1}{in} e^{-inx} \right]_0^{\pi} = \frac{1}{2\pi} \left( -\frac{1}{in} ((-1)^n - 1) \right) \\ = -\frac{1}{2\pi in} ((-1)^n - 1)$$

$$f(x) = \frac{1}{2} + \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} -\frac{1}{2\pi in} ((-1)^n - 1) e^{inx}$$

$$f(x) = \frac{1}{2} + \sum_{n \text{ odd}} -\frac{1}{\pi in} e^{inx}$$

## PROBLEM 1

$$(jii) \quad f(x) = \begin{cases} -\frac{\pi}{2} & -\pi < x < -\pi/2 \\ x & -\pi/2 < x < \pi/2 \\ \frac{\pi}{2} & \pi/2 < x < \pi \end{cases}$$

$$d_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0 \quad \leftarrow \text{Net Area under curve is zero.}$$

$$d_n = \frac{1}{2\pi} \left[ \underbrace{\int_{-\pi}^{-\pi/2} -\frac{\pi}{2} e^{-inx} dx}_{(1)} + \underbrace{\int_{-\pi/2}^{\pi/2} x e^{-inx} dx}_{(2)} + \underbrace{\int_{\pi/2}^{\pi} \frac{\pi}{2} e^{-inx} dx}_{(3)} \right]$$

$$(1) = -\frac{\pi}{2} \left( -\frac{1}{in} e^{-inx} \right) \Big|_{-\pi}^{-\pi/2} = \frac{\pi}{2in} \left( e^{i\frac{\pi}{2}n} - e^{i\pi n} \right) = \frac{\pi}{2in} (i^n - (-1)^n)$$

$$(2) \quad \int_{-\pi/2}^{\pi/2} x e^{-inx} dx$$

$u = x$   
 $du = dx$

$v = -\frac{1}{in} e^{-inx}$   
 $dv = e^{-inx} dx$

$$= -\frac{x}{in} e^{-inx} + \frac{1}{in} \int e^{-inx} dx \\ = -\frac{x}{in} e^{-inx} + \frac{1}{n^2} e^{-inx} \Big|_{-\pi/2}^{\pi/2} = \left( -\frac{\pi}{2in} (-i)^n + \frac{1}{n^2} (-i)^n \right) - \left( \frac{\pi}{2in} i^n + \frac{1}{n^2} i^n \right)$$

$$(3) = \frac{\pi}{2} \left( -\frac{1}{in} e^{-inx} \right) \Big|_{\pi/2}^{\pi} = -\frac{\pi}{2in} ((-1)^n - i^n)$$

$$d_n = \frac{1}{2\pi} \left[ \frac{\pi}{in} (i^n - (-1)^n) - \frac{\pi}{2in} ((-i)^n + i^n) + \frac{1}{n^2} ((-i)^n - i^n) \right]$$

$$f(x) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} d_n e^{inx} \quad \text{where } d_n \text{ is as above}$$

## PROBLEM 2

$$(a) \sum_{n=-\infty}^{\infty} d_n e^{inx} = \underbrace{\dots + d_{-3}e^{-i3x} + d_{-2}e^{-i2x} + d_{-1}e^{-ix}}_{\sum_{n=-\infty}^{-1} d_n e^{inx}} + d_0 + \underbrace{d_1 e^{ix} + d_2 e^{i2x} + d_3 e^{i3x} + \dots}_{\sum_{n=1}^{\infty} d_n e^{inx}}$$

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$$(b) d_0 + \sum_{n=-\infty}^{-1} d_n e^{inx} + \sum_{n=1}^{\infty} d_n e^{inx}$$

$$= d_0 + \sum_{n=-\infty}^{-1} d_{-n} e^{-inx} + \sum_{n=1}^{\infty} d_n e^{inx}$$

$$= d_0 + \sum_{n=1}^{\infty} d_{-n} e^{-inx} + \sum_{n=1}^{\infty} d_n e^{inx}$$

$$= d_0 + \sum_{n=1}^{\infty} d_{-n} (\cos(-nx) + i \sin(-nx)) + \sum_{n=1}^{\infty} d_n (\cos(nx) + i \sin(nx))$$

$$= d_0 + \sum_{n=1}^{\infty} d_{-n} \cos(nx) - i d_{-n} \sin(nx) + d_n \cos(nx) + i d_n \sin(nx)$$

$$= d_0 + \sum_{n=1}^{\infty} (d_n + d_{-n}) \cos(nx) + i (d_n - d_{-n}) \sin(nx)$$

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$$(c) d_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = a_0 \quad \checkmark$$

$$\begin{aligned} d_n + d_{-n} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx + \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) (2 \cos(nx)) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx \\ &= a_n \quad \checkmark \end{aligned}$$

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$$i(d_n - d_{-n}) = \frac{i}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx - \frac{i}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx$$

$$= \frac{i}{2\pi} \int_{-\pi}^{\pi} f(x) (e^{-inx} - e^{inx}) dx$$

$$= \frac{i}{2\pi} \int_{-\pi}^{\pi} f(x) (-2i \sin(nx)) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = b_n \quad \checkmark$$